

CASE 41

PEMPHIGUS VULGARIS

Autoimmune attack on the integrity of the skin.

Autoimmune disease results when the adaptive immune response is directed against self antigens. The particular pathology of the disease depends on the nature of the self antigen and the type of immune response that is mounted against it. In the type II autoimmune diseases (see Fig. 40.1), the pathology is caused by autoantibodies that interact with self antigens in the extracellular matrix or on cell surfaces. In myasthenia gravis, for example, autoantibodies against the acetylcholine receptor on skeletal muscle block its function at the neuromuscular junction and cause paralysis, whereas in Graves' disease an autoantibody against the receptor for thyroid-stimulating hormone acts as an agonist, stimulating receptor activity and causing hyperthyroidism. In the type II autoimmune disease pemphigus vulgaris, the self-reactive agent is an autoantibody against a structural protein of the epidermal cells of the skin. Its actions result in the skin cells coming apart from each other; the affected skin blisters and is destroyed.

'Pemphigus' is derived from the Latin word for blister, and 'vulgaris' means common or ordinary. Patients with this disease have autoantibodies against desmoglein-3, which is a protein component of the desmosome—one of the intercellular junctions that link skin cells and other epithelial cells tightly to each other. Desmogleins are members of the cadherin family of cell adhesion molecules, proteins that effect intercellular adhesion in a calcium-dependent manner. Disruption of desmoglein causes blisters to form in the skin and on the mucous membranes; extensive sloughing of the skin may ensue (Fig. 41.1).

Evidence that pemphigus vulgaris is caused by an autoantibody comes from the observation that infants born to mothers with the disease have a transient period of skin blistering during the neonatal period. There is also a good animal model in which human IgG from patients with pemphigus injected into neonatal mice reproduces pemphigus.



Fig. 41.1 Lesions of pemphigus vulgaris that have coalesced to produce a large plaque. The plaque has a moist surface that is oozing and has crusts of dried serum from blister fluid. Photograph courtesy of Razzaque Ahmed.

TOPICS BEARING ON THIS CASE:

Peptide binding by MHC class II antigens

Humoral autoimmunity

Epitope spreading

HLA associations with disease

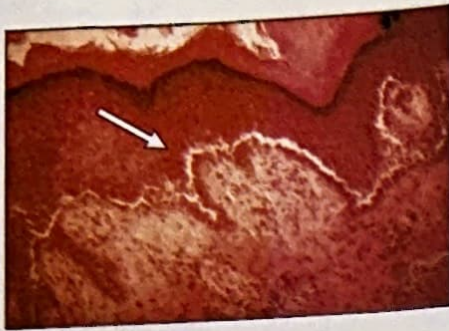


Fig. 41.2 Histopathology of an early skin lesion of pemphigus vulgaris. Stained with hematoxylin and eosin. An epidermal, suprabasal vesicle (blister) is clearly seen. Photograph courtesy of Razzaque Ahmed.

Persistent skin sore and ulceration of mouth; resistant to usual therapies; do biopsy.

Diagnosis of pemphigus vulgaris; start immunosuppressive therapy.

Condition not responding to steroids.

Responding to cyclophosphamide; decrease dose gradually.

The case of Arthur Sammler, Esq.: a hoarse lawyer with a sore mouth and blistering skin.

Arthur Sammler is a lawyer of Ashkenazi Jewish descent. He was 55 years of age when an irritation in his throat became persistent enough for him to consult a physician. The physician suggested that he gargle with warm saline, which seemed to help. A month later, however, after a long day in court he became very hoarse and the hoarseness persisted for a week. A week later he noticed a 'sore' on his right cheek. He told the physician he was under considerable stress over a custody case, and the physician diagnosed a herpes simplex infection and advised him to take oral acyclovir. The lesion did not improve.

At a more thorough examination 2 weeks later, numerous erosions and ulcers in the mucosa of his mouth and gums were also revealed. These were thought to result from a yeast infection with *Candida albicans* and he was treated with a Mycostatin (nystatin) mouthwash.

However, new lesions soon appeared on the palate, uvula, and tongue, and Mr Sammler returned to the physician. This time, a biopsy of the lesions was taken by an oral surgeon. While awaiting the results of the biopsy, Mr Sammler developed erosions of the skin on his scalp and neck.

Microscopic examination of the skin biopsy revealed disruption of the epidermal layer of the skin characteristic of pemphigus vulgaris (Fig. 41.2). Mr Sammler was sent to a dermatologist, who took another biopsy for immunofluorescence studies; blood samples were taken for analysis. The immunofluorescence studies revealed deposits of IgG in an intercellular pattern within the epidermis (Fig. 41.3). His serum contained an antibody against desmoglein-3 (known commonly as 'pemphigus antibody') in a titer of 1:640.

Mr Sammler was treated with large doses of prednisone, 120 mg per day. After 3 weeks of this therapy, during which time no improvement occurred, new lesions appeared on his back and scalp. The dose of prednisone was increased to 180 mg per day. Two weeks later he developed a persistent cough, fever, and chest pain. A chest radiograph during an emergency room visit suggested that he had developed interstitial pneumonia due to *Pneumocystis jirovecii*. He was admitted to hospital and treated for this infection.

It was found that his pemphigus antibody titer had risen to 1:1280. He was started on cyclophosphamide at 150 mg per day, and was told to decrease his prednisone dose by 10 mg every 10 days and to rinse his mouth daily with hydrogen peroxide and elixir of Decadron (a corticosteroid). New lesions kept appearing, so the cyclophosphamide was increased to 200 mg per day.

Over the next 8 weeks the pemphigus antibody titer decreased to 1:320; no new lesions appeared, and the older ones healed. The cyclophosphamide dose was decreased over an 8-month period, and Mr Sammler remained symptom-free after cyclophosphamide was stopped. In the year since then, he has noticed the occasional new lesion after damage to his skin; these new lesions are promptly

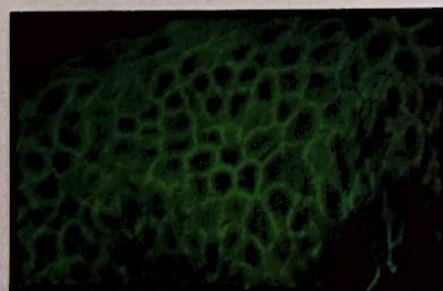


Fig. 41.3 Direct immunofluorescence study of tissue around a pemphigus vulgaris lesion. The normal-appearing tissue around the lesion contains deposits of IgG, other immunoglobulins, and complement components in the intercellular spaces of the entire stratum malpighii of the epidermis. Photograph courtesy of Razzaque Ahmed.

injected with a corticosteroid. His pemphigus antibody titer has remained at 1:80 during this asymptomatic period.

Pemphigus vulgaris.

The case of Arthur Sammler illustrates how an autoimmune disease that was frequently fatal in the past can now be controlled by immunosuppressive drugs. The epidermis, with its cells held tightly together, forms an important barrier to the entry of pathogens into the body. Extensive blistering and sloughing of skin destroy this barrier; in the past, this often resulted in bloodstream infection (septicemia) with *Staphylococcus aureus*, a common contaminant of human skin. Treatment with cyclophosphamide or non-steroidal immunosuppressive drugs often eliminates the symptoms, thus allowing skin to heal and reducing the chances of opportunistic infections. Newer methods of treatment, such as rituximab (an anti-CD20 antibody that depletes B cells), have shown promise in patients who do not respond to corticosteroids in combination with immunosuppressive agents.

The mechanism of autoimmune tissue destruction seems to be somewhat unusual in pemphigus vulgaris. In some autoimmune diseases, the autoantibodies cause tissue destruction by stimulating the complement system, but the autoantibody in pemphigus vulgaris is of the IgG4 subclass and does not fix complement. Some other mechanism must therefore be invoked to explain the disruption of desmoglein-3 adhesion. It is thought that binding of the autoantibody to its antigen in some way causes upregulation of serine proteinase activity on the surface of the epidermal cells, and that this results in the proteolytic digestion of desmoglein-3. The particular enzyme that digests the desmoglein-3 is as yet unknown.

Long-term study of the antibody response in some patients with pemphigus vulgaris has shown that the response evolves to involve epitopes other than the one against which antibodies are initially directed. This is known as epitope spreading and has been found to correlate with the clinical progression of disease. Antibodies present before the onset of disease are specific for epitopes in the extracellular part of the molecule nearest to the cell membrane. These antibodies do not bind to cell-surface desmoglein, nor do they transfer disease when injected into mice. With disease onset, antibodies specific for the domains farthest away from the cell surface become detectable (Fig. 41.4). These IgG antibodies bind to cell-surface desmoglein, and when injected into mice, they induce disease. The process by which the immune response initially targets epitopes in one part of an antigenic molecule and then progresses to other, non-cross-reactive, epitopes of the same molecule is called intra-molecular epitope spreading. In pemphigus, it is only with epitope spreading that disease-causing antibodies are made.

Pemphigus vulgaris is encountered most frequently in Ashkenazi Jews and people of Mediterranean descent. Studies have shown strong association with an HLA haplotype found mainly in Ashkenazi Jew ethnic group. This HLA haplotype includes HLA-DR4 and HLA-DQ3, which are present in virtually all Ashkenazi Jews with pemphigus vulgaris. Patients are usually heterozygous for this haplotype and only in rare cases are they homozygous. Unaffected relatives of patients with pemphigus who bear the same HLA haplotype frequently also have antibodies against desmoglein-3. However, these antibodies are of the IgG1 subclass and do not cause disease. In other ethnic groups, pemphigus vulgaris is usually associated with HLA-DR14, HLA-DRw6, and HLA-DQ5.

A molecular explanation has been discovered for the association of pemphigus vulgaris with HLA-DR4. Each major histocompatibility complex (MHC) class II variant can bind a subset of all possible peptides; the actual peptides that can be bound by any given MHC molecule are determined by the particular pattern of amino-acid binding sites in the peptide-binding cleft of the MHC molecule (Fig. 41.5). The β chain of the MHC class II molecule DR is designated DRB1, and

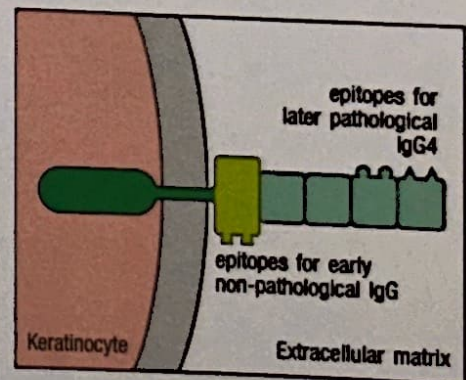
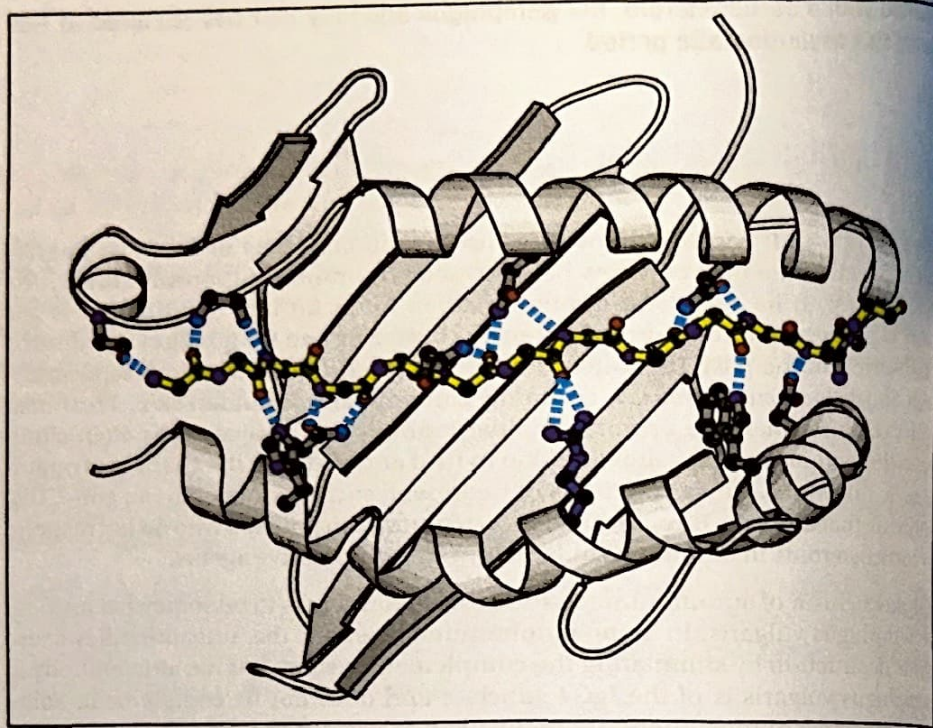


Fig. 41.4 The progressive involvement of desmoglein epitopes in pemphigus vulgaris. This skin-blistering disease is caused by antibodies that bind to desmoglein-3, an adhesion molecule in the cell junctions that hold keratinocytes together. When the autoimmune B-cell response begins, harmless antibodies are made against the extracellular region of desmoglein-3 nearest to the cell surface. In time, the response spreads and antibodies are made against the region farthest away from the cell surface. These antibodies cause disease and are of the IgG4 subclass.

Fig. 41.5 Peptides bind to MHC class II molecules by interactions along the length of the binding groove. A peptide (yellow; shown as the peptide backbone only, with the amino terminus to the left and the carboxy terminus to the right), is bound by an MHC class II molecule through a series of hydrogen bonds (dotted blue lines) that are distributed along the length of the peptide. The hydrogen bonds toward the amino terminus of the peptide are made with the backbone of the class II polypeptide chain, whereas throughout the peptide's length, bonds are made with residues that are highly conserved in MHC class II molecules. The side chains of these residues are shown in gray on the ribbon diagram of the MHC class II binding groove. Most DR4 alleles have a positively charged arginine (shown here in purple) or lysine at position 71 of the β chain. The DRB1*0402 allele that is linked to susceptibility to pemphigus has a negatively charged glutamic acid residue instead.



22 known subtypes of this chain have been identified by DNA typing among those bearing DR4 (DRB1*04). Only one of these variants—DRB1*0402—is associated with pemphigus vulgaris. It differs from the other 21 variants at amino acid position 71, which lies in one of the binding pockets in the peptide-binding groove of the MHC molecule. In DRB1*0402, position 71 is a negatively charged glutamic acid, whereas all the other variants have positively charged amino acid residues (lysine or arginine) at this position. Only DRB1*0402 will bind the antigenic peptide derived from desmoglein-3 and present it to T lymphocytes.

Questions.

- 1 The intravenous administration of gamma globulin frequently improves autoimmune disease. It has been observed that the administration of gamma globulin to patients with pemphigus vulgaris significantly lowers the titer of IgG4 anti-desmoglein-3 and improves the clinical course of the disease. How do you explain this effect?
- 2 Relatives of patients with pemphigus vulgaris who share the DRB1*0402 MHC class II type have IgG1 antibodies against desmoglein-3. These antibodies fix complement but do not cause disease. Can you speculate why this antibody does not cause disease, whereas the IgG4 antibody causes disease?
- 3 What might explain the difference between unaffected people who make only the IgG1 antibody against desmoglein-3 and the patients who make the IgG4 antibody? Can you design an experiment to test your hypothesis?
- 4 We have seen cyclophosphamide used to eradicate bone marrow cells in Cases 8 and 27. Why is it useful in the treatment of pemphigus vulgaris?
- 5 Why did Mr Sammler develop pneumonia caused by *Pneumocystis jirovecii* during his initial treatment for pemphigus vulgaris?